THE DECOMPOSITION, INERTIA AND WILD INERTIA GROUPS OF A MAXIMAL IDEAL RESTRICT SURJECTIVELY ALONG A GALOIS SUBTOWER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span contains two commutative diagrams and the definition of e' and e; all are delivered verbatim. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Decomposition, inertia and wild inertia groups of a maximal ideal in a Galois extension of a discretely valued field. fixed at statement lines 33624-33632. Are the groups whose restriction behaviour is being described.
- Normality of the residue field extension and the lifting of automorphisms of a normal subextension. used at proof lines 33663-33668. Produces the right vertical arrow in the first diagram and its commutativity.
- The inertia character into the roots of unity, and the relation between uniformizers and ramification indices. used at proof lines 33677-33688. Is the computation that gives commutativity of the second diagram up to the power e'/e.

Source reference. Original lemma 0BME: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14627–14627. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 33224: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33224–33224. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 32869: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32869–32869. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let A be a discrete valuation ring with fraction field K. Let $M/L/K$ be a tower with M/K and L/K finite Galois. Let C, B be the integral closure of A in M, L. Let $\mathfrak{m}' \subset C$ be a maximal ideal and set $\mathfrak{m} = \mathfrak{m}' \cap B$. Let*

$$P \subset I \subset D \subset Gal(L/K) \quad and \quad P' \subset I' \subset D' \subset Gal(M/K)$$

be the wild inertia, inertia, decomposition group of $\mathfrak{m}$ and $\mathfrak{m}'$. Then the canonical surjection $Gal(M/K) \to Gal(L/K)$ induces surjections $P' \to P$, $I' \to I$, and $D' \to D$. Moreover these fit into commutative diagrams

$$\begin{array}{ccc} D' & \longrightarrow & Aut(\kappa(\mathfrak{m}')/\kappa_A) \\ \downarrow & & \downarrow \\ D & \longrightarrow & Aut(\kappa(\mathfrak{m})/\kappa_A) \end{array} \quad and \quad \begin{array}{ccc} I' & \xrightarrow[\theta']{} & \mu_{e'}(\kappa(\mathfrak{m}')) \\ \downarrow & & \downarrow{\scriptstyle (-)^{e'/e}} \\ I & \xrightarrow{\theta} & \mu_e(\kappa(\mathfrak{m})) \end{array}$$

where e' and e are the ramification indices of $A \to C_{\mathfrak{m}'}$ and $A \to B_{\mathfrak{m}}$.

Proof. The fact that under the map $\mathrm{Gal}(M/K) \to \mathrm{Gal}(L/K)$ the groups P', I', D' map into P, I, D is immediate from the definitions of these groups. The commutativity of the first diagram is clear (observe that since $\kappa(\mathfrak{m})/\kappa_A$ is normal every automorphism of $\kappa(\mathfrak{m}')$ over κ_A indeed induces an automorphism of $\kappa(\mathfrak{m})$ over κ_A and hence we obtain the right vertical arrow in the first diagram, see Lemma 33224 and Fields, Lemma 0BME).

The maps $I' \to I$ and $D' \to D$ are surjective by Lemma 32869. The surjectivity of $P' \to P$ follows as P' and P are p-Sylow subgroups of I' and I.

To see the commutativity of the second diagram we choose a uniformizer π' of $C_{\mathfrak{m}'}$ and a uniformizer π of $B_{\mathfrak{m}}$. Then $\pi = c'(\pi')^{e'/e}$ for some unit c' of $C_{\mathfrak{m}'}$. For $\sigma' \in I'$ the image $\sigma \in I$ is simply the restriction of σ' to L. Write $\sigma'(\pi') = c\pi'$ for a unit $c \in C_{\mathfrak{m}'}$ and write $\sigma(\pi) = b\pi$ for a unit b of $B_{\mathfrak{m}}$. Then $\sigma'(\pi) = b\pi$ and we obtain

$$b\pi = \sigma'(\pi) = \sigma'(c'(\pi')^{e'/e}) = \sigma'(c')c^{e'/e}(\pi')^{e'/e} = \frac{\sigma'(c')}{c'}c^{e'/e}\pi$$

As $\sigma' \in I'$ we see that b and $c^{e'/e}$ have the same image in the residue field which proves what we want. $\square$